

Research update

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Probability Weight Matrix

PWM represents position specific nucleotide preference

Inputs - protein binding sequences

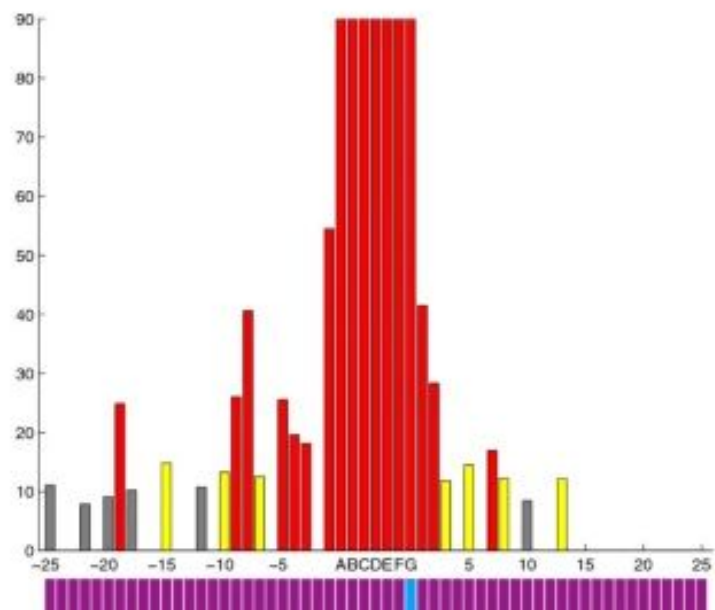
Output - model that can predict how well a protein will bind to a sequence

Limitations

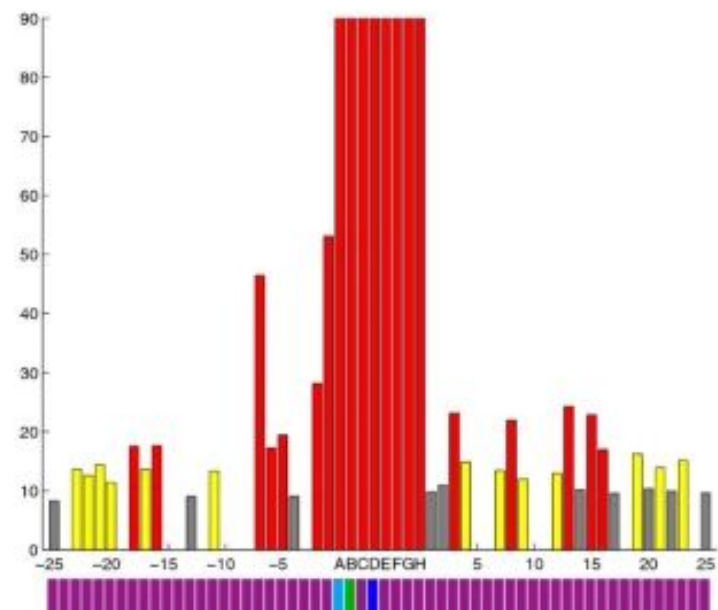
- Assumes nucleotides are independent
- Can only represent a continuous binding site (core-motif)

Strigham paper identifies conserved sequences that PWM miss.

HB



KNI



MARZ algorithm

MARZ ranks common nucleotide patterns in binding areas

- Can capture nucleotide interdependence
- Allows for gaps in sequences, can be expanded past core-motif

Inputs

- known TF binding sequences
- background sequences
- ChIP seq peak data

MARZ

- Creates a PWM from the known sequences and background sequences.
- Labels strong and weak binding sequences across the genome (ChIP seq data).
- Breaks down a known sequence into chunks, from a single base to the entire sequence (n-mers).
- Scores the n-mers with the PWM.
- Combines the chunks with gaps of ignored nucleotides to create gapped n-mers.
- The n-mers and gapped n-mers are scored based on how often they appear in strong and weak binding sites vs the background sequences.

Outputs

- The frequency of the gapped n-mer in known binding site sequences.
- The expected frequency of finding the gapped n-mer in a random sequence.
- Identifies most interesting gapped n-mers.

Validation

- Check how often gapped n-mer show up near significant TF binding locations.

Zellers found dependency between 1st and 6th nucleotide “mkkkkm”, outside of core-motif.

What should i do?

FTZ/EVE Chimera paper

- Run MARZ on aligned sequences of FTZ and EVE sites from the chimera paper, look for significant nucleotide patterns

Limitations

- Only a few sequences for each, might overfit

What should i do?

Strigham paper, Flanking sequences of TF binding sites

Run MARZ on the sequences identified in the Strigham paper with conserved flanks.

Limitations

- MARZ is computationally heavy, larger sequence could be tough
- The binding sequences are predicted